

## Chapter 2: Health and Diseases

Textbook Exercise Solutions &amp; Answers



### 1. Distinguish between- Infectious and non – infectious diseases.

| Infectious diseases                                                                                                                                                     | Non-infectious diseases                                                                                                                                                                                            |
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| 1. The diseases which can be transmitted from one patient to other person are called infectious diseases.                                                               | 1. The diseases which cannot be transmitted from one patient to other person, are called non-infectious diseases.                                                                                                  |
| 2. Contaminated air, water, food or carrier! vector animals or insects spread the infectious diseases.                                                                  | 2. Non-infectious diseases are not transmitted through any medium.                                                                                                                                                 |
| 3. infectious diseases are acute diseases as they are suddenly caused due to infections from either bacteria, virus or protozoan. Thus, their symptoms are seen sooner. | 3. Some non-infectious diseases are due to hereditary causes, some are a clue to failure in metabolism, some due to deficiency of the nutrients. Their symptoms are not seen suddenly but are developed gradually. |
| 4. Antibiotics or antihelminth medicines are used to treat the infectious diseases. E.g. Tuberculosis, hepatitis, leprosy, cough and cold, etc.                         | 4. Antibiotics cannot be used for treating non-infectious diseases. These diseases have a proper treatment. E.g. Diabetes, cancer, heart diseases, etc. Tuberculosis, hepatitis, leprosy, cough and cold, etc.     |

### 2. Identify the odd term.

(a) Malaria, hepatitis, elephantiasis, dengue.

Hepatitis (All other diseases are caused by the carrier mosquito.)

(b) Plague, AIDS, Cholera, T.B.

AIDS [All others are bacterial (caused by bacteria) diseases.]

### 3. Answer in one to two sentences.

(a) Which are various media of spreading the infectious diseases?

Infectious diseases spread through contaminated air, water, food, vectors such as insects, animals and man.

**(b) Give the names of five non-infectious diseases other than given in the lesson.**

Asthma, cataract, diseases of kidney such as kidney stones and renal failure, arthritis, Alzheimer which is a condition during old age, hypertension, migraine, etc.

**(c) Which are the main reasons of diabetes and heart diseases?**

Improper lifestyle, wrong type of diet, lack of exercise, excessive mental stress and strain, imbalance in secretion of hormones, etc. are the main reasons for diabetes and heart diseases.

#### **4. What can be achieved/can be prevented?**

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**(a) Drinking boiled and filtered water.**

1. When water is boiled, all the disease causing pathogens present in it are killed.
2. Different diseases like cholera, enteritis, diarrhoea, dysentery, hepatitis, typhoid are caused by such water-borne pathogens.
3. If we boil the water we are protected against all such diseases.
4. If the i water is filtered we can avoid infections from nematode *Dracunculus* which causes *Naru*. By boiling and filtering water, even the epidemics by such infectious diseases can be controlled.

**(b) Avoiding smoking and alcoholism.**

1. Smoking and alcoholism are two dangerous habits that cause addictions. Alcoholism causes disorders of liver.
2. The addict suffers from malnourishment.
3. His I mental and psychological conditions become abnormal. There is social and familial) impacts too due to alcoholistn.
4. Smoking is an invitation for cancer. In cigarettes/ bidis, there is hazardous nicotine. Nicotine is not only toxic but is also carcinogenic. Cancers of oral cavity, tongue, respiratory tract, lungs is very common among smokers. Therefore, addictions like smoking and alcoholism should always be avoided.

**(c) Regular balanced diet and exercise.**

1. Regular and balanced diet results into perfect health. The disease fighting power or immunity of a person is increased due to healthy diet.
2. One can avoid of frequent infections and also mental well-being is established.
3. Heart disease, diabetes, obesity, etc. can be avoided by not consuming high caloric junk food.
4. Exercise improves blood circulation. Many disorders which are caused by faulty lifestyle can be prevented by having regular balanced diet and exercise, thus one must always follow these for well-being.

**(d) Proper checking of blood before blood donation.**

1. There are definite blood groups.
2. During blood transfusion, the donor's and recipient's blood should be well-matched with each other. Otherwise, blood gets clumped inside the body of recipient.
3. Through infected blood, the viruses such as those causing hepatitis B or AIDS are transmitted to other healthy persons. Thus, for prevention of transmission of such diseases, blood should be checked before blood donation.

**5. Read the passage and answer the questions.**

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*Master X' is a 3-year-old child. He is living with his family in a slum. The public toilet is present near his house. His father is drunkard. His mother does not know the importance of balanced diet.*

**(a) Master 'X' can suffer from which different possible diseases in above conditions?**

Master 'X' stays in an area which is devoid of cleanliness. The public toilet is near his house. This indicates that he will have to fight against many infectious diseases. Because his father is a drunkard, there will always be a dearth of money in the house. He may not be able to afford, enough and balanced food. Thus his diet must be deficient in vitamins and minerals.

Moreover, his mother is ignorant about the importance of health and a balanced diet. This must be causing Master 'X' malnourishment and loss of immunity. The financial conditions and the addition of the father must be causing stress in the house.

This will further add stress on Master 'X' resulting into susceptibility to infections. As it is due to the location the germs are around and thus they may be attacking Master 'X'. He will on and off suffer from digestive disorders! such as typhoid, hepatitis, cholera, enteritis, etc.

**(b) How will you help him and his family in this situation?**

Initially, we shall help Master 'X' to procure healthy and balanced diet having fruits, vegetables, milk, etc. His mother will be taught importance of the balanced diet. The surroundings should be clean and hygienic. Food should be covered, houseflies should not be allowed to contaminate the food. Germicidal floor cleaners and disinfectants should be used to keep off the houseflies and cockroaches. His father will be taken to de-addiction centres such as 'Alcoholics Anonymous'. He will be persuaded to leave alcoholism.

**(c) Which disease can occur to the father of master 'X'?**

Father of Master 'X' can develop diseases of liver and kidney.

## 6. Give the preventive measures of following diseases

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### (a) Dengue

Dengue is transmitted through bite of mosquito of *Aedes aegypti*. DEN-1, 2 virus belonging to the type – flavivirus is the causative pathogen for Dengue. Wherever there is stagnant or accumulated water, there is possibility of mosquito breeding. Therefore, care is to be taken to drain of such water. Thus this is a very important preventive measure. Especially, in the man-made containers and in clean water, the *Aedes* mosquito prefers to breed.

Therefore, such water storages should be either covered or should be decanted. Another way to keep off from dengue is to increase our immunity to fight against the virus. There is vaccine called CYD-TDV or Dengvaxia which is synthesised in 2017. But it is still not considered to be completely safe.

### (b) Cancer

The most important preventive measure is to remain away from the carcinogenic substances. Tobacco, gutkha, cigarette, bidi, etc. are addictive things which are very bad for our health. The nicotine present in these cause cancers of oral cavity and of respiratory system. Radiations can also cause cancer.

We should not expose ourselves to hazardous radiations. Balanced and healthy diet, proper exercise and living stress free with mental balance are some of the preventive measures of the cancer. Only for few types of cancer like cervical cancer in women, vaccine has been developed.

### (c) AIDS

When blood transfusion is done, the blood should be checked for the presence of HIV. The used syringes, needles, etc. should not be used without sterilization. Through blood and blood products and body fluids HIV finds its way into the body. Therefore, these precautions should be taken. Unsafe sexual contact is the most important mode of transmission of HIV. Thus one should never deal with such dangerous acts. Awareness about AIDS and HIV is the real preventive measure.

## 7. Explain the importance.

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### (a) Balanced diet.

The diet that contains all the nutrients in the balanced proportion is called balanced diet. One can avoid malnutrition by taking balanced diet. The immunity increases due to balanced diet containing good proportion of vitamins and minerals. Some diseases can be avoided due to raised immunity. Wrong lifestyle and wrong diet can cause hypertension, diabetes or heart disease. To maintain our health and keep the body in equilibrium, we have to take balanced diet.

**(b) Physical exercise/Yogasanas.**

By exercise and yoga, the blood circulation of body is improved. Body and joints remain flexible. Mental strain and stress is reduced. Insomnia (inability to fall asleep), arthritis, indigestion, and some other disorders can be avoided. The person who performs exercise, always remains away from the addictions. Yoga makes levels of hormones, enzymes, etc. in equilibrium. By keeping control over breathing through pranayama many respiratory and circulatory disorders can be prevented.

**8. Make a list.**

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**(a) Viral diseases.**

AIDS, Hepatitis, Influenza, Rabies, Polio.

**(b) Bacterial diseases.**

Typhoid, Tuberculosis, Cholera, Leprosy.

**(c) Diseases spread through insects.**

Malaria, Dengue, Elephantiasis.

**(d) Hereditary diseases.**

Diabetes, Haemophilia, Muscular dystrophy, Colourblindness.

**9. Write the information on modern diagnostics and treatments of cancer.**

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1. Following methods are used as diagnostic methods to detect the cancer. Techniques like CT scan, MRI scan, mammography, biopsy.

Treatment of the cancer is done by the following methods:

2. For treatment of cancer, some conventional methods are used. Along with these methods, chemotherapy, radiation therapy and surgery are commonly used to treat cancerous growth and tumours. New and modern techniques of robotic and laparoscopic surgery are also used for the treatment.

**10. Enlist the names and composition of the medicines present at your home.**

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Students are expected to enlist medicines present at their home along with their composition and uses (e.g., Paracetamol – for fever and mild pain; Cetirizine – antihistamine for allergic reactions; ORS – oral rehydration salts for dehydration; Antacids – for hyperacidity).